

Project 4. Graph Algorithms

June 5, 2026

Question 1: Return Correlation

Bounds on ρ_{ij} . The quantity ρ_{ij} is the Pearson correlation coefficient between the log-normalized return time series of stocks i and j , defined as

$$\rho_{ij} = \frac{\langle r_i(t) r_j(t) \rangle - \langle r_i(t) \rangle \langle r_j(t) \rangle}{\sqrt{(\langle r_i(t)^2 \rangle - \langle r_i(t) \rangle^2)(\langle r_j(t)^2 \rangle - \langle r_j(t) \rangle^2)}}.$$

By the Cauchy–Schwarz inequality, the Pearson correlation coefficient always satisfies

$$-1 \leq \rho_{ij} \leq +1.$$

The boundary values are achieved as follows:

- $\rho_{ij} = +1$: the two return series are perfectly positively linearly related, meaning the stocks move in exactly the same direction and in the same proportions at every time step.
- $\rho_{ij} = -1$: the two return series are perfectly negatively linearly related, moving in exactly opposite directions.
- $\rho_{ij} = 0$: there is no linear correlation between the two return series.

Justification for log-normalized returns. We use the log-normalized return

$$r_i(t) = \log(1 + q_i(t))$$

rather than the simple return $q_i(t)$ for the following reasons.

1. **Approximate normality.** For small $q_i(t)$, the approximation $\log(1 + q) \approx q$ holds, but the log transform brings the empirical distribution of returns closer to a Gaussian. Since the Pearson correlation coefficient has the most reliable statistical properties under the normality assumption, using $r_i(t)$ yields more robust correlation estimates.
2. **Time-additivity.** Multi-period log returns are additive over time:

$$\log \frac{p_i(T)}{p_i(0)} = \sum_{t=1}^T r_i(t).$$

Simple returns, by contrast, must be multiplied across periods, making aggregation over different time horizons inconsistent.

3. **Symmetry.** Log returns are symmetric: a price increase by a factor k (log return = $\ln k$) is exactly reversed by multiplying back by $1/k$ (log return = $-\ln k$), so the two cancel precisely. For example, a +50% gain ($\times 1.5$, log return = $+0.405$) followed by a return to the original price ($\times \frac{2}{3}$, i.e. -33.3% , log return = -0.405) sums to zero. With simple returns, the same round trip appears as +50% and -33.3% , which look asymmetric — illustrating why log returns are the natural choice.
4. **Variance stabilisation.** The log transform compresses the influence of extreme price movements and reduces heteroscedasticity in the return series, producing a more stationary signal and making the temporal average $\langle \cdot \rangle$ a more meaningful summary statistic.

Question 2: Constructing Correlation Graphs

We compute the edge weight $w_{ij} = \sqrt{2(1 - \rho_{ij})}$ for every pair of the 495 stocks using the daily log-normalized return time series. To ensure reliable correlation estimates, we retain only stocks with at least 700 trading days of history, reducing the universe from 505 to 495 stocks. After aligning on common dates, the return matrix spans 701 trading days.

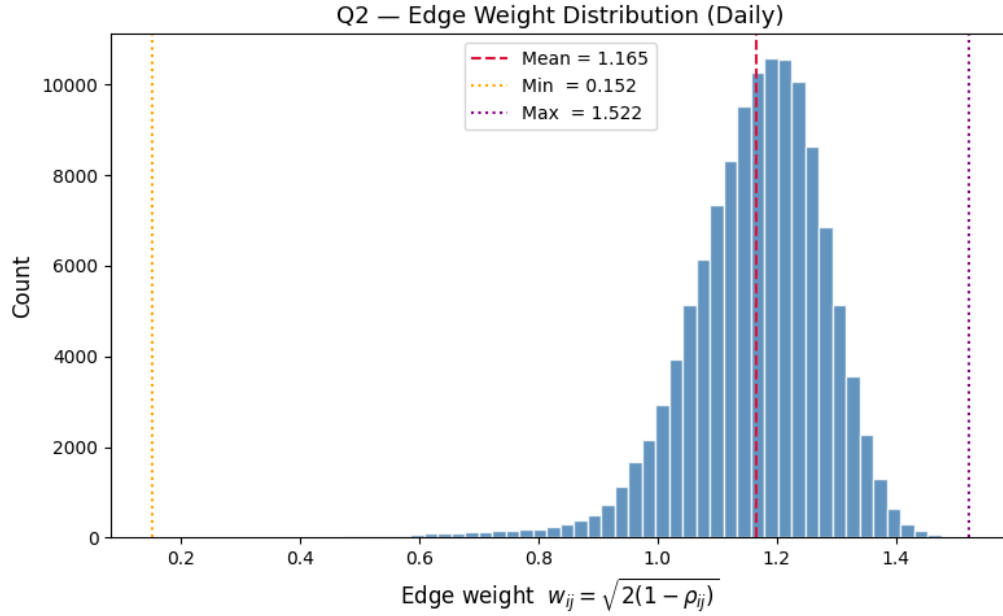


Figure 1: Un-normalized distribution of edge weights w_{ij} for the daily correlation graph (495 stocks, 701 trading days).

Statistic	Value
Minimum	0.1516
Maximum	1.5215
Mean	1.1651
Std. dev.	0.1161

Table 1: Summary statistics of the daily edge weight distribution.

The distribution is left-skewed and unimodal, with the bulk of edge weights concentrated between 1.0 and 1.3. A weight of $w_{ij} = 0$ corresponds to $\rho_{ij} = 1$ (perfect positive correlation), $w_{ij} = \sqrt{2} \approx 1.414$ to $\rho_{ij} = 0$ (no correlation), and $w_{ij} = 2$ to $\rho_{ij} = -1$ (perfect negative correlation). The observed mean of 1.165 indicates that the average stock pair has a moderate positive correlation ($\rho_{ij} \approx 0.32$), consistent with the broad market factor driving most S&P 500 constituents in the same direction. The minimum weight of 0.152 corresponds to $\rho_{ij} \approx 0.988$, reflecting nearly co-integrated pairs such as dual-class shares or close industry peers.

Question 3: Minimum Spanning Tree (MST)

We extract the MST of the daily correlation graph using Kruskal's algorithm, which retains the $|V| - 1 = 494$ edges of lowest weight (highest correlation) among the 495 stocks with sufficient history. The resulting MST has 495 connected nodes and 494 edges. The highest-degree node is **ITW** (Illinois Tool Works, Industrials sector) with degree 19, acting as the central hub bridging multiple sector clusters.

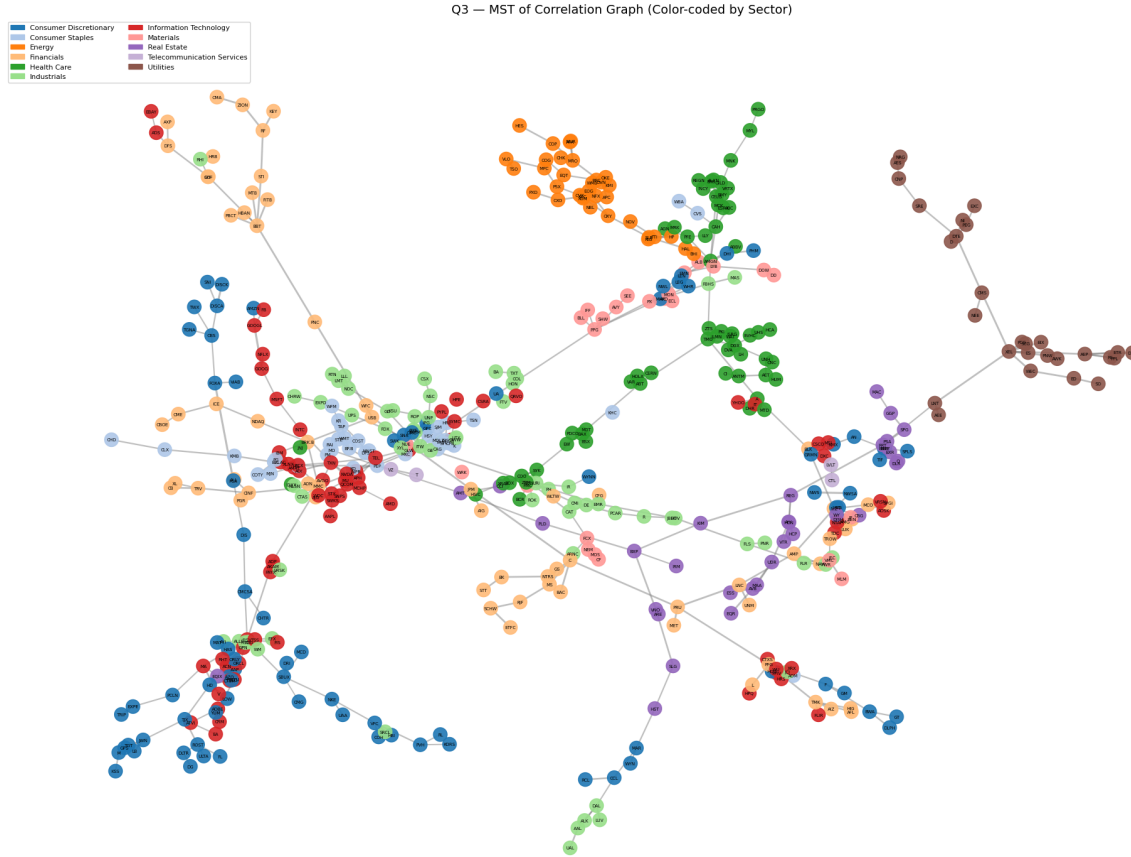


Figure 2: MST of the daily correlation graph, color-coded by market sector. Each node represents a stock and each edge connects the pair with the highest correlation (lowest w_{ij}) in the spanning tree.

Vine cluster pattern. The MST exhibits a clear *vine cluster* structure: stocks belonging to the same sector consistently cluster into elongated, branch-like sub-trees. This pattern arises because intra-sector correlations are systematically stronger than inter-sector correlations — stocks in the same sector are driven by common economic factors (e.g., oil prices for Energy, interest rates for Financials, consumer spending for Consumer Discretionary), so the MST preferentially connects them first.

Several vine clusters are clearly visible in Figure 2:

- **Energy** (orange): forms a large, elongated vine, reflecting the strong common dependence on oil prices across the sector.
- **Utilities** (brown): forms a compact, isolated cluster connected to the rest of the tree through a single bridge edge, consistent with utilities being driven by regulatory and interest-rate factors largely decoupled from the broader market.
- **Consumer Discretionary** (blue): occupies a broad vine extending across one side of the tree.
- **Real Estate** (purple): forms a separate cluster, consistent with the sector’s sensitivity to interest rates rather than broad market movements.
- **Information Technology** (red): more dispersed than other sectors, reflecting the internal heterogeneity of the sector (hardware, software, semiconductors).

Hub nodes such as ITW (degree 19) are predominantly large-cap diversified companies that serve as bridges between sector clusters, confirming that the MST encodes meaningful economic relationships. Note that the absolute positions of clusters in the figure depend on the spring layout algorithm and carry no geographic meaning; only the relative groupings and connectivity are interpretable.

Question 4: Community Detection on the MST

We apply the Walktrap algorithm (step size = 4) to the daily MST. The algorithm detects **53 communities**. The resulting partition is compared side-by-side against the true sector labels in Figure 3.



Figure 3: Left: 53 Walktrap communities detected on the daily MST. Right: true sector labels for comparison.

Metric	Value
Number of communities	53
Homogeneity	0.7229
Completeness	0.4653

Table 2: Walktrap community detection metrics for the daily MST.

Interpretation. Homogeneity measures whether each detected community contains only members of a single true sector; completeness measures whether all members of a true sector are assigned to the same community. A homogeneity of 0.72 indicates that most communities are relatively pure — the stocks within a given community tend to belong to the same sector. The lower completeness of 0.47 reflects the fact that Walktrap over-partitions the tree: because the MST vine clusters are long and linear, the random-walk process splits each sector into several smaller communities rather than recovering it as a single unit. This is an inherent limitation of applying graph-based community detection to trees, where every edge is a bridge and the algorithm has no dense subgraph structure to anchor larger communities.

Question 5: Sector Clustering in the MST

We evaluate the sector predictability of the MST using the metric

$$\alpha = \frac{1}{|V|} \sum_{v_i \in V} P(v_i \in S_i),$$

under two definitions of $P(v_i \in S_i)$:

- **Method 1 (MST neighbors):** $P(v_i \in S_i) = |Q_i|/|N_i|$, where Q_i is the set of neighbors of node i in the MST that belong to the same sector, and N_i is the full neighbor set.
- **Method 2 (random baseline):** $P(v_i \in S_i) = |S_i|/|V|$, i.e., the fraction of all stocks that belong to sector S_i , representing the probability of a randomly chosen stock being in the correct sector.

Method	α
MST neighbors	0.8441
Random baseline	0.1146
Ratio	$7.37\times$

Table 3: Alpha metric for the daily MST under two definitions of $P(v_i \in S_i)$.

Interpretation. The MST-neighbor α of 0.84 means that, on average, 84% of a stock’s MST neighbors belong to the same sector. The random baseline of 0.11 reflects the sector composition of the full dataset (11 sectors of varying sizes). The ratio of $7.37\times$ shows that an MST neighbor is more than 7 times more likely to share the same sector than a randomly selected stock, confirming that the MST captures strong sector structure and would be a useful tool for predicting the sector of an unknown stock from its neighbors.

Question 6: Correlation Graphs for Weekly Data

We repeat Questions 2–5 using weekly data sampled on Mondays, yielding 130 observations per stock after alignment on common dates.

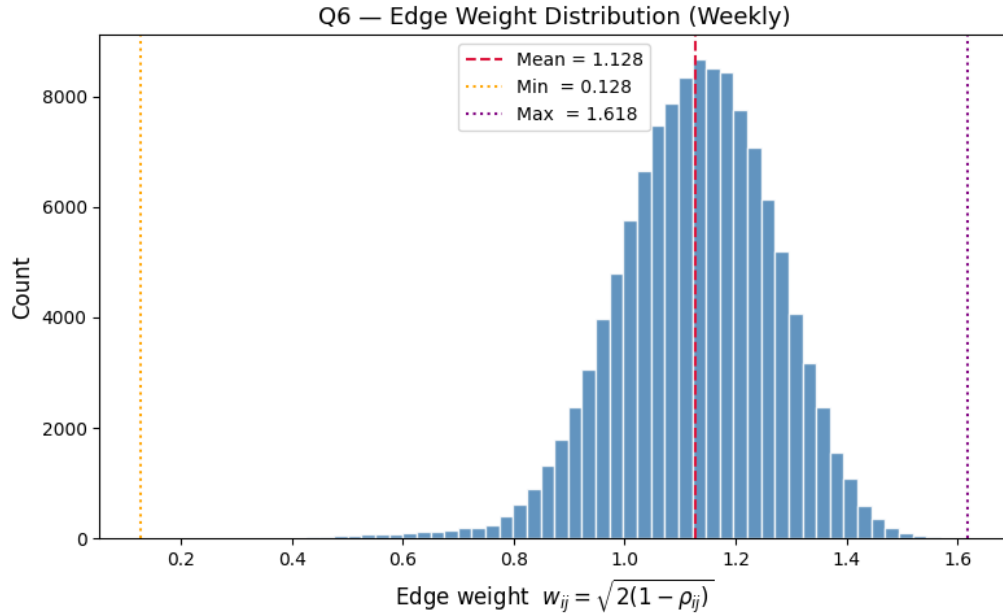


Figure 4: Un-normalized distribution of edge weights w_{ij} for the weekly correlation graph.

Statistic	Daily	Weekly
Minimum	0.1516	0.1280
Maximum	1.5215	1.6183
Mean	1.1651	1.1275
Std. dev.	0.1161	0.1424

Table 4: Edge weight statistics: daily vs. weekly.

Q2 (Weekly) — Edge Weight Distribution. Compared to daily data, the weekly distribution is wider (std. dev. 0.142 vs. 0.116), indicating that weekly correlation estimates are more dispersed. This is expected: with 130 weekly observations versus 701 daily, the Pearson estimator carries higher sampling variance, and the weekly sampling interval captures a different frequency band of co-movement than daily data, reducing the sharpness of intra-sector correlation differences.

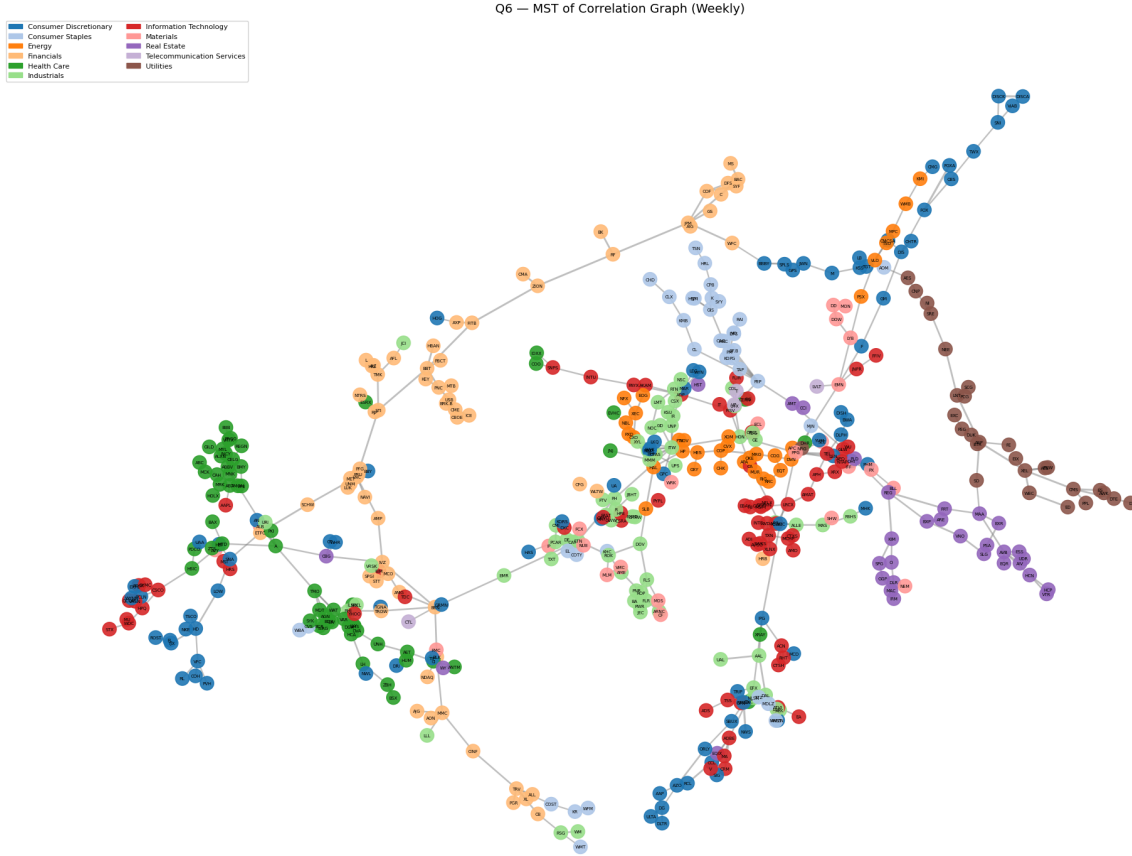


Figure 5: MST of the weekly correlation graph, color-coded by sector. Highest-degree hub: EFX (Industrials, degree 11).

Q3 (Weekly) — MST. The weekly MST retains the vine cluster structure observed in the daily MST, with same-sector stocks grouping into elongated branches. The hub node shifts from ITW (Industrials, degree 19) in the daily MST to EFX (Industrials, degree 11), and the maximum degree drops, indicating a flatter, less star-like tree. Sector clusters are less compact, reflecting the noisier weekly correlation estimates.

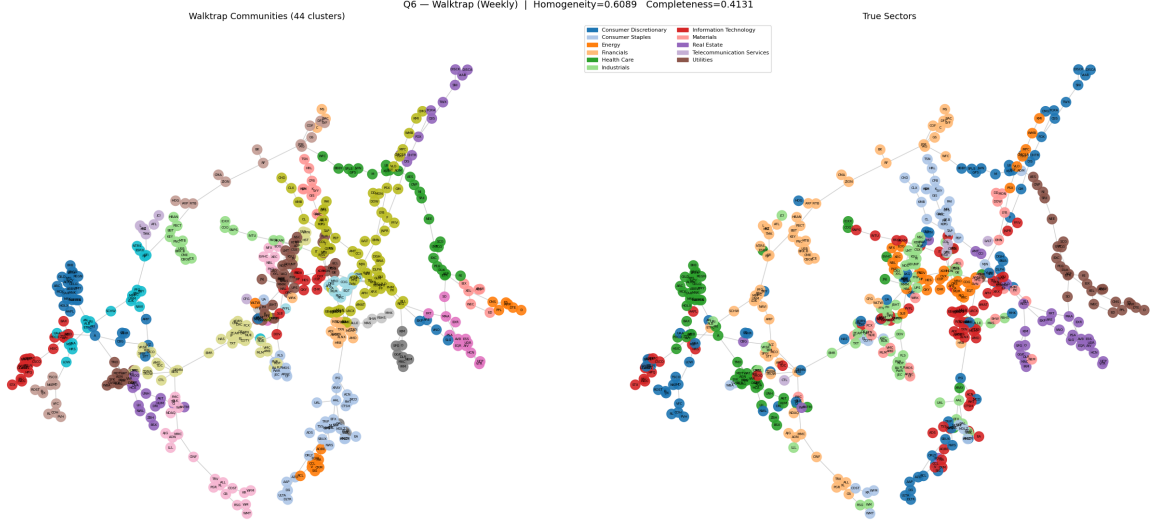


Figure 6: Left: 44 Walktrap communities on the weekly MST. Right: true sector labels. Homogeneity = 0.6089, Completeness = 0.4131.

Metric	Daily	Weekly
Communities	53	44
Homogeneity	0.7229	0.6089
Completeness	0.4653	0.4131

Table 5: Walktrap metrics: daily vs. weekly MST.

Q4 (Weekly) — Walktrap Community Detection. Both homogeneity and completeness drop from daily to weekly, confirming that community detection becomes less reliable with fewer observations. With 130 weekly data points, correlation estimates carry higher variance, causing the MST to include more spurious cross-sector edges that fragment the vine clusters and reduce the alignment between detected communities and true sectors.

Method	Daily	Weekly
MST neighbors	0.8441	0.7661
Random baseline	0.1146	0.1146
Ratio	$7.37\times$	$6.69\times$

Table 6: Alpha metric: daily vs. weekly MST.

Q5 (Weekly) — Alpha Metric. The MST-neighbor α falls from 0.84 (daily) to 0.77 (weekly), and the ratio drops from $7.37\times$ to $6.69\times$. An MST neighbor is still more than 6 times more likely to share the same sector than a randomly chosen stock, but the higher-variance weekly correlations reduce the sector-predictive power of the tree.

Question 7: Correlation Graphs for Monthly Data

We repeat Questions 2–5 using monthly data sampled on the 15th of each month. Per the project specification, months where the 15th falls on a non-trading day are skipped entirely. After applying this rule and aligning on common dates, we obtain 22 observations per stock.

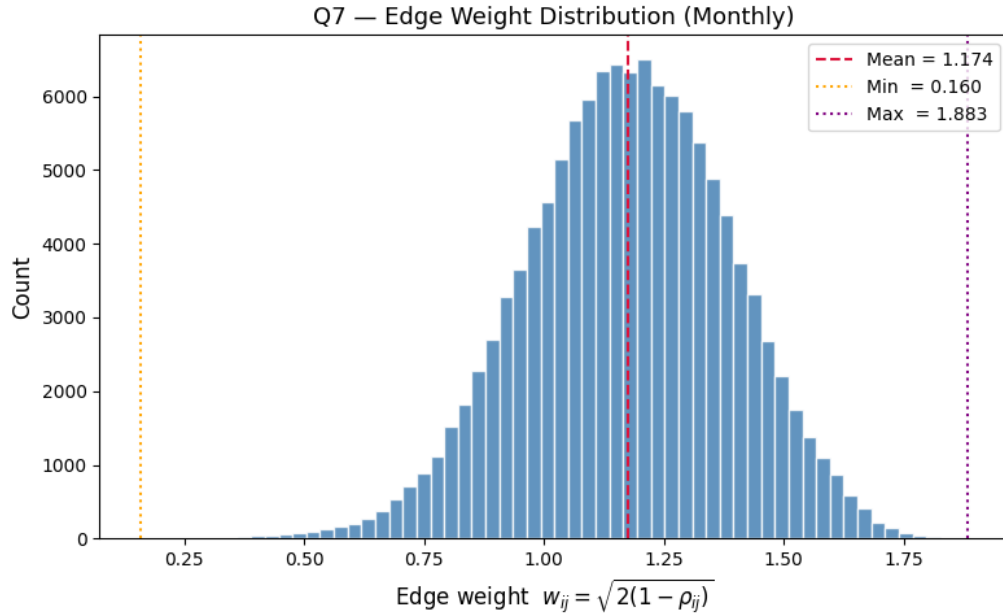


Figure 7: Un-normalized distribution of edge weights w_{ij} for the monthly correlation graph.

Statistic	Daily	Weekly	Monthly
Minimum	0.1516	0.1280	0.1602
Maximum	1.5215	1.6183	1.8832
Mean	1.1651	1.1275	1.1741
Std. dev.	0.1161	0.1424	0.2130

Table 7: Edge weight statistics across all three granularities.

Q2 (Monthly) — Edge Weight Distribution. The monthly distribution is substantially wider than the daily and weekly distributions (std. dev. 0.213 vs. 0.116), spanning a broader range but still retaining a visible peak around 1.1–1.3. With only 22 monthly observations, the Pearson correlation estimates carry higher sampling variance, allowing more pairs to reach extreme weight values by chance. However, the distribution is not uniform — the mean of 1.174 is consistent with a moderate positive correlation signal, indicating that 22 months is sufficient to preserve some economic structure, unlike what would arise from only a handful of observations.

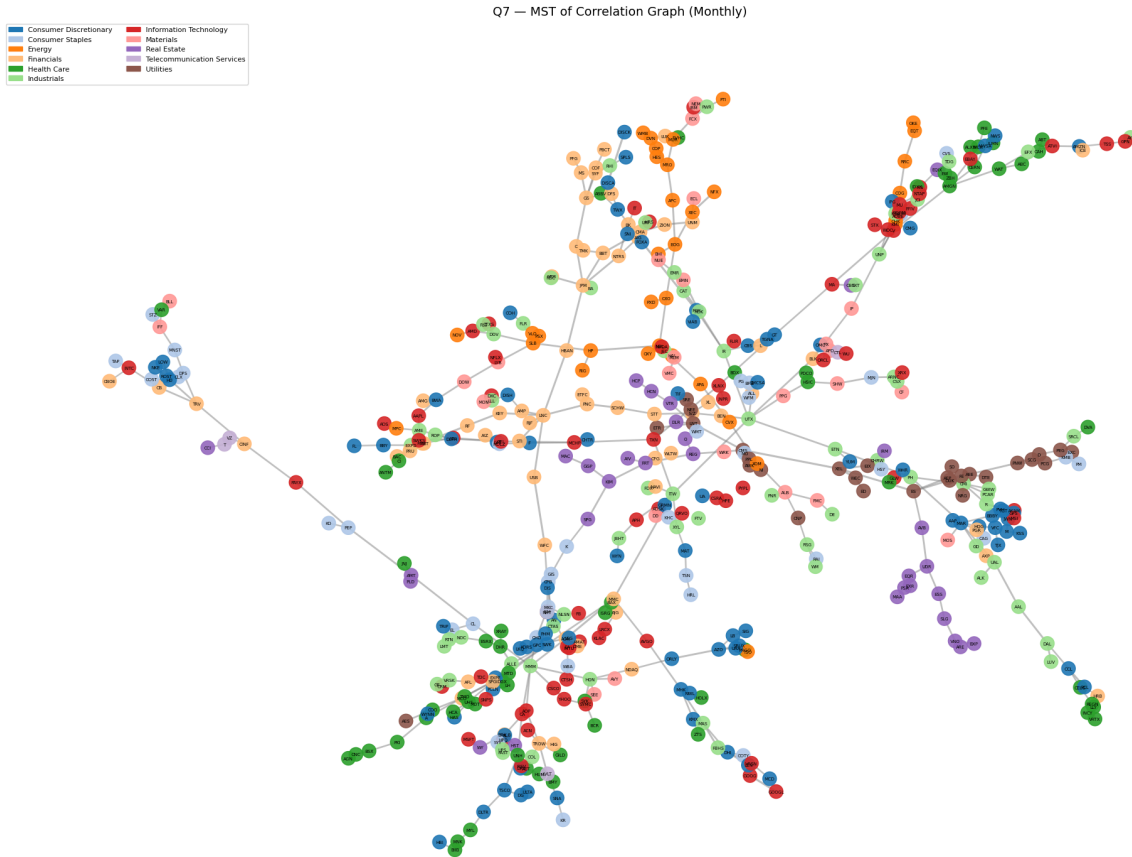


Figure 8: MST of the monthly correlation graph, color-coded by sector. Highest-degree hub: ITW (Industrials, degree 8).

Q3 (Monthly) — MST. The monthly MST shows a more diffuse structure compared to the daily and weekly MSTs, with sector clusters less sharply defined. The hub node is ITW (Illinois Tool Works, Industrials) with degree 8, lower than both the daily (degree 19) and weekly (degree 11) cases, indicating a flatter tree. Same-sector groupings are still partially visible but with more cross-sector mixing than in finer granularities.

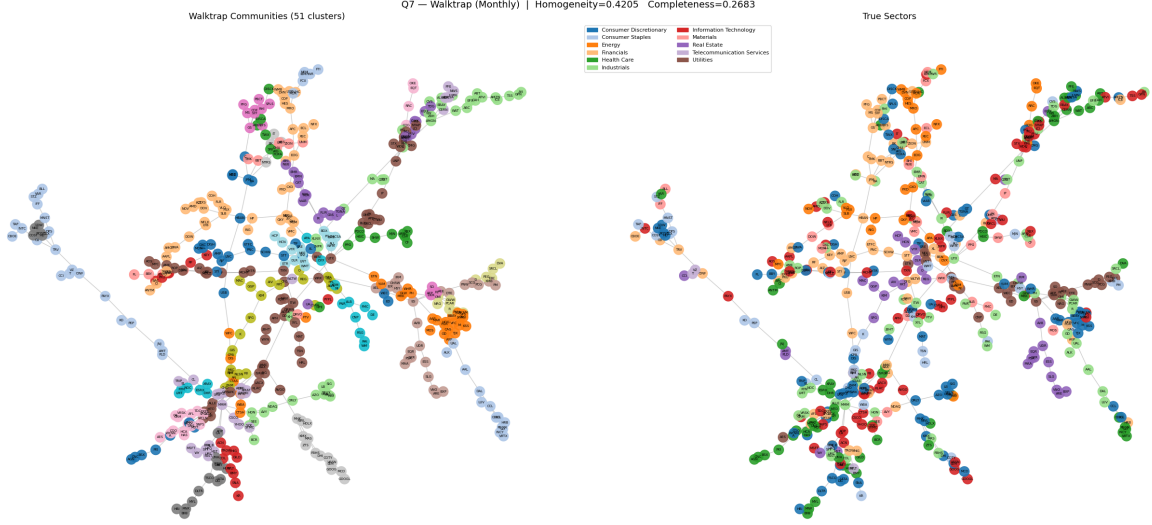


Figure 9: Left: 51 Walktrap communities on the monthly MST. Right: true sector labels. Homogeneity = 0.4205, Completeness = 0.2683.

Metric	Daily	Weekly	Monthly
Communities	53	44	51
Homogeneity	0.7229	0.6089	0.4205
Completeness	0.4653	0.4131	0.2683

Table 8: Walktrap metrics across all three granularities.

Q4 (Monthly) — Walktrap Community Detection. Both homogeneity and completeness reach their lowest values at monthly granularity, confirming that community detection degrades as the number of observations decreases. The 51 detected communities show weaker alignment with true sectors, reflecting the higher-variance correlation estimates from 22 monthly data points.

Method	Daily	Weekly	Monthly
MST neighbors	0.8441	0.7661	0.4832
Random baseline	0.1146	0.1146	0.1146
Ratio	7.37×	6.69×	4.22×

Table 9: Alpha metric across all three granularities.

Q5 (Monthly) — Alpha Metric. The MST-neighbor α drops to 0.48 at monthly granularity, with a ratio of 4.22× over the random baseline. While this is notably lower than daily (7.37×) and weekly (6.69×), it remains well above chance, indicating that 22 monthly observations still preserve meaningful sector structure in the MST, though with reduced predictive reliability compared to finer granularities.

Question 8: Comparison of Daily, Weekly, and Monthly Data

Table 10 consolidates all metrics computed in Questions 2–5 across the three temporal granularities.

Metric	Daily	Weekly	Monthly
Observations per stock	701	130	22
Edge weight mean	1.1651	1.1275	1.1741
Edge weight std. dev.	0.1161	0.1424	0.2130
Walktrap communities	53	44	51
Homogeneity	0.7229	0.6089	0.4205
Completeness	0.4653	0.4131	0.2683
α (MST neighbors)	0.8441	0.7661	0.4832
α (random baseline)	0.1146	0.1146	0.1146
α ratio	7.37×	6.69×	4.22×

Table 10: Summary of all metrics across daily, weekly, and monthly granularities.

What changes across granularities? Every quality metric — homogeneity, completeness, α , and α ratio — decreases monotonically as the sampling frequency decreases from daily to weekly to monthly. The edge weight standard deviation increases from 0.116 (daily) to 0.213 (monthly), reflecting increasingly unreliable correlation estimates as the number of observations shrinks. The monthly edge weight distribution is substantially wider than the daily, with a broader peak and a longer right tail, indicating that with only 22 data points the Pearson estimator has high variance and allows more pairs to reach extreme weight values by chance. The MST topology also changes: the daily hub (ITW, degree 19) gives way to a lower-degree hub at weekly (EFX, degree 11) and an even flatter tree at monthly (ITW, degree 8), consistent with correlation structure becoming less concentrated as observations decrease.

What remains similar? The α ratio remains well above 1 at all three granularities, confirming that MST neighbors are always more likely to share the same sector than a randomly

chosen stock. The broad vine cluster pattern is partially visible even at monthly granularity. The mean edge weight stays close to 1.13–1.17 across all granularities, suggesting that the average pairwise correlation level is relatively stable; it is the *reliability* of individual estimates, not their average, that degrades with coarser sampling.

Which granularity gives the best results for sector prediction? Daily data gives the best results by a clear margin, evidenced by the highest α ratio ($7.37\times$), homogeneity (0.72), and completeness (0.47). With 701 observations, the Pearson correlation estimates have low variance, producing a reliable MST that faithfully reflects true sector structure. Weekly data is a moderate compromise: 130 observations yield noticeably weaker but still meaningful sector clustering (α ratio $6.69\times$). Monthly data with 22 observations performs worst across all metrics (α ratio $4.22\times$), though it still captures some sector signal above the random baseline.

The fundamental reason is statistical: the standard error of the Pearson correlation estimator scales as $(1 - \rho^2)/\sqrt{T}$. For $T = 22$ (monthly), this is approximately 0.20, large enough to blur the distinction between intra- and inter-sector pairs. Increasing T to 701 (daily) reduces this error by a factor of $\sqrt{701/22} \approx 5.6$, yielding estimates that reliably rank intra-sector pairs above inter-sector pairs and produce meaningful MST and community structure.